

IN AN EMERGENCY

111

the only number a student needs to know

A **LAN-only campus emergency & internal calling system** built on Asterisk/FreePBX. It runs entirely over campus Wi-Fi with **no internet, no SIM card, and no cloud** — so it keeps working when normal networks are down. Deployable as a fixed campus install **or** a self-powered disaster-response van with corner repeaters.

WHAT IT IS

Asterisk PBX inside a QEMU Ubuntu VM on a Windows laptop. Android phones (Linphone) register as SIP softphones over the local LAN. One-command installer stands up the whole stack.

IDENTITY MODEL

SAP ID = extension = username. Responder roles are **positions (4xxx)** staffed by shift — never a personal account. Same person + different job = same ID, different context.

WHY LAN-ONLY

Survives internet / cellular / cloud / PSTN failure — the entire point. **111 is human-first** and never depends on AI, internet, or a mobile signal.

THE SYSTEM IN ONE PICTURE

Android (Linphone) on campus Wi-Fi → AP → Switch/Router → Asterisk / FreePBX

dial 111

↓

ert_emergency_queue (press 1 → first-aid) → offline coach (102) if no answer point is free. ERT Lead + backup alerted in the background, in parallel.
(every 111 call is recorded from the first second)

NUMBERS TO REMEMBER

111	Campus emergency hotline
102	Offline panic-coach (auto fallback)
199	Drill / test line
*22 / *23	Responder shift on / off
700s	Emergency paging · 9000s conference

ALSO IN THE BOX

- **Operations Console** (:8080) + two always-on **LED-TV wallboards** (safety / ops)
- **Shared CardDAV phonebook** pushed to phones — dial responders by name
- **Multi-language voice prompts** (Piper neural TTS): English + Hindi, Telugu, Malayalam, Urdu, Nepali
- **Missed-emergency follow-up** callback workflow; nothing is dropped silently
- Stable name `upes-ecs.local` (mDNS) — no phone reconfig on a network change

STATUS SNAPSHOT — validated live on real Android phones + real SAP-ID accounts

DONE

SIP registration, two-way audio, 111 rings a real answering phone, offline coach, recording, status API, console, docs & config.

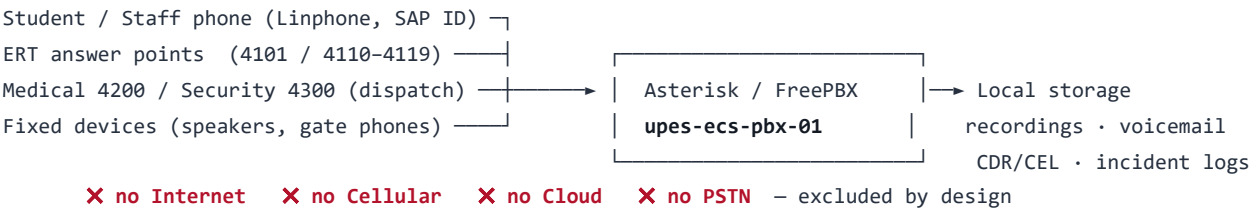
BUILDABLE NEXT

Real recorded prompts, QR phone provisioning, Console served over the LAN, filled test-evidence sheet.

NEEDS HARDWARE / DECISION

Move off QEMU NAT to a real LAN IP (**#1**), TLS+SRTP, always-on ERT phones, van UPS, full pilot & go-live.

SYSTEM CONTEXT — WHO TALKS TO WHAT (LAN-ONLY)



INSIDE THE PBX — COMPONENTS

COMPONENT	ROLE
chan_pjsip	SIP registration & signalling; enforces auth, LAN-only, per-endpoint context
Dialplan / contexts	The brain — routes 111/199, grants only what each role may reach
app_queue	ert_emergency_queue — ring-all available ERT positions, 20s
MixMonitor	Records the whole 111/199 call → WAV linked to an incident ID
app_voicemail	Emergency voicemail when all responders miss
app_confbridge	Incident-command rooms 9000–9004 (9000 recorded when active)
app_page	Live paging to zones 700–799 (PIN on all-campus)
CDR / CEL	Call detail + event logs; carry EMERGENCY / DRILL labels

HELPER SCRIPTS (/OPT/UPES-ECS)

- incident_id.sh — mints ERT-YYYYMMDD-NNNN
- missed_incident.sh — records an unanswered emergency
- upes-ecs-healthcheck.sh — live health status
- retention-cleanup.sh — 90-day purge of audio
- mass_callout.sh — roll-call / mass call-out (EAS)
- shift_event.sh · ert-shift.sh — queue join/leave audit
- log_access_denied / log_paging / log_conf — audit trails

LIVE CONTROL PLANE

upes_api.py (FastAPI, :8090 in the VM) exposes GET /status + whitelisted POST /exec . The Operations Console proxies /api/* over an SSH tunnel and drives the wallboards.

IDENTITY VS PERMISSION — TWO ORTHOGONAL IDEAS

WHO YOU ARE (NUMBER)	WHAT YOU MAY DO (CONTEXT)
Humans → SAP ID (9-digit student / 8-digit staff)	ctx_student · ctx_staff
Responder positions → 4xxx	ctx_ert · ctx_ert_lead · ctx_responder · ctx_control_room
Fixed devices → 4700s	ctx_fixed_device
Service codes → 111/199/700s/9000s	ctx_ai_helpline · ctx_admin

Positions, not people → continuity through shift handover, no crisis-time provisioning. Least privilege: emergency controls locked to emergency roles.

DEPLOYMENT MODES

- Mode A — Campus (fixed): PBX on a campus server on mains + UPS. The everyday emergency line.
- Mode B — Mobile / Van: PBX-in-a-van on generator + battery + solar, with corner repeaters. For disasters, off-grid, or as failover for the campus PBX.
- Same config, same numbers, same SOP across both modes. Multi-campus (Bidholi ↔ Kandoli) via a rooftop wireless bridge.

HIGH-AVAILABILITY OPTION

2× Jetson Nano (native ARM) active/standby via keepalived VIP + Juniper voice-VLAN/QoS.

DESIGN PRINCIPLES — THE "WHY"

- LAN-only — survives every external failure
- 111 is human-first — AI never gates it
- Positions, not people — shift continuity
- Record + log everything on 111; student calls stay private
- Least privilege by context
- Fixed answer points so answering never depends on one battery

EMERGENCY & SERVICE CODES	
#	NAME / BEHAVIOUR
111	Campus Emergency Hotline — the one number. Human-first; rings the ERT queue; recorded.
102	Offline Panic-Coach — deterministic first-aid (CPR / bleeding / choking / fire / lockdown), zero internet. Auto fallback when no human answers.
101	Local-first AI triage (Ollama + Whisper + Piper, no cloud). Always escalates to 111. Never dialed by a caller.
199	Drill / test — simulates 111, no real dispatch, logged <code>DRILL-ONLY</code>
198	Echo / audio test — checks mic & speaker
196	Internal AI test line
100	Deprecated & removed — softphones hijack 100 as "Police". Not routed or dialable.

FEATURE CODES		
CODE	ACTION	WHO
*22 / *23	Shift on / off (join/leave 111 queue)	Responders
*45 / *46	Pause / resume within a shift	ERT
*77	Silent SOS / duress → pages ERT zone	Everyone
*9	One-tap incident bridge → room 9000	ERT
*80<ext>	Intercom / auto-answer	ERT · admin
105	Request a callback	Students · staff
411	Dial-by-name directory	Staff
211-215	Dept hunt: Security / Medical / Warden / Ops / IT	Staff
720-723	Announce: evacuate / avoid / all-clear / assemble	ERT-Lead

RESERVED RANGES	
RANGE	USE
SAP ID	Every real person — extension = username
4101	ERT Lead / Incident Commander
4110-4119	ERT Operator positions — answer the 111 queue
4120	ERT Control Room
4200-4299	Medical — 4200 dispatch, 4201-02 seats
4300-4399	Security — 4300 dispatch, 4301 Lead
4400-4499	Warden / Hostel
4500-4599	Admin / Operations
4600-4699	IT / Network
4700-4799	Fixed devices — speakers, gate phones
700-799	Emergency paging zones
9000-9099	Incident-command conference rooms

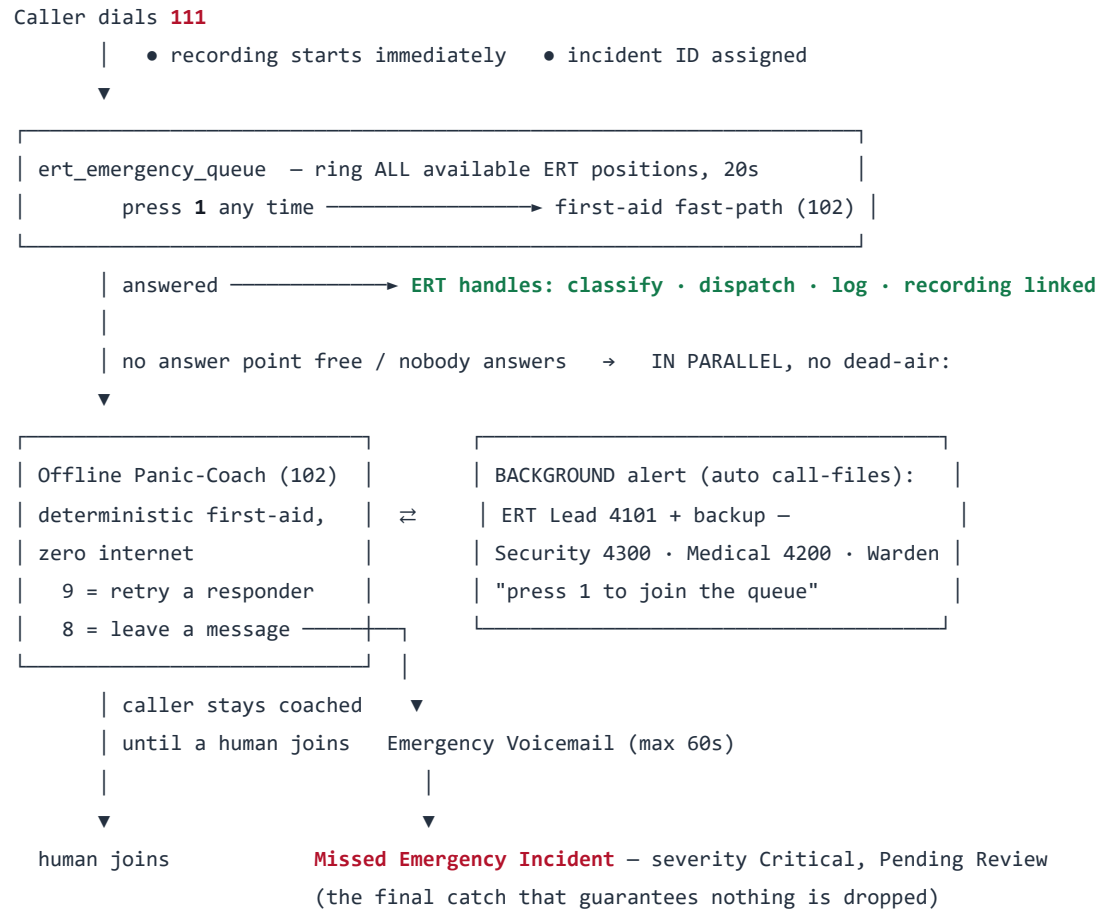
PAGING ZONES 700-799 · RESTRICTED + LOGGED	
700	All-campus broadcast — ERT Lead only, PIN
701-705	Academic · Hostels · Gates · Medical · Admin

CONFERENCE ROOMS 9000-9099 · PIN, RESPONDER-ONLY	
9000	Main Incident Command (recorded, max 20)
9001-9004	Security · Medical · Warden · Ops rooms (max 10)

IDENTITY & LOGGING FORMATS		
Incident ID	Recording file	Call-log labels
ERT-20260704-0001	ERT-..._500120597_...wav	EMERGENCY_111_CALL · DRILL-ONLY

The 111 Emergency Call Path

NO DEAD-AIR, NO SERIAL RINGING — THE CALLER IS COACHED THE WHOLE TIME



QUEUE RULES

- Strategy = **ring all available**
- Skip busy / paused / offline
- Healthy = **min 2** available responders
- Coach + background alert fire the moment no answer point is free

RECORDED & RETAINED

- Every **111 / 199** call recorded from second one
- Recording linked to the incident ID
- **Student-to-student calls are never recorded**
- 9000 recorded only when a real incident is active

IN-COACH CONTROLS

- **Press 1** — first-aid fast-path (any time)
- **9** — retry reaching a responder
- **8** — leave a voicemail message

GOLDEN RULES

- **111 is human-first** — never depends on AI, internet, or cellular; the offline coach (102) is the automatic fallback.
- **Unsure? Escalate.** Never drop an emergency call silently.
- **Only 111 / 199 are recorded.** Student-to-student calls never are.
- **Emergency controls are role-restricted** — paging, conference, recordings.
- **Back up before every change. Test with 199 before 111.**

RESPONDER POSITIONS — STAFFED BY SHIFT, NOT BY A NAMED PERSON

EXT	ROLE	CONTEXT	ANSWERS 111 QUEUE?
4101	ERT Lead / Incident Commander	ctx_ert_lead	Escalation / background-alert target
4110-4119	ERT Operator positions (desks + reserves)	ctx_ert	YES — queue member
4120	ERT Control Room	ctx_control_room	YES
4200s	Medical — 4200 dispatch + 2 seats	ctx_responder	Dispatch target
4300s	Security — 4300 dispatch, 4301 Lead	ctx_responder	Dispatch target
4400s	Warden / Hostel	ctx_responder	Dispatch target
4500s	Admin / Operations	ctx_responder	Dispatch target
4600s	IT / Network	ctx_responder	Dispatch target
4700s	Fixed devices — speakers, gate phones	ctx_fixed_device	No — unstaffed

ERT positions answer the 111 queue; Medical / Security / Warden / Ops / IT are dispatch targets, not queue answerers. Each department has a dispatch front-door (round number) + minimal answer seats.

STORAGE & RETENTION

DATA	KEEP
Emergency recordings & voicemail	90 days
Incident / missed records	1 year
CDR / CEL, queue / paging / conf logs	1 year
Config (git + FreePBX backup)	30 daily + 12 weekly

No captured call audio is ever left on a server or shipped in an installer — only generated TTS prompts. Final legal retention TBD (UPES admin).

PERMISSION CONTEXTS

- ctx_student
 - ctx_staff
 - ctx_ert
 - ctx_ert_lead
 - ctx_responder
- ctx_responder_lead
 - ctx_control_room
 - ctx_ai_helpline
 - ctx_fixed_device
 - ctx_admin

Include chain: staff ⊃ student, ERT ⊃ staff. A code added to ctx_student (SOS *77, callback 105) reaches everyone; one added to ctx_staff reaches staff + responders but not students.

COORDINATION FEATURES

- Mass call-out / roll-call (EAS)** — rings a group, plays a message, reads "1 = safe" and tallies a report. Shown as "UPES-EAS" <111> , never anonymous.
 - Pre-recorded announcements** 720-723 to all-campus speakers (evacuate / avoid / all-clear / assemble).
 - Silent SOS *77** — raises a CRITICAL incident with minimal talk-path and pages the ERT zone.
- Intercom *80** — hands-free auto-answer for ERT / admin.
 - Callback 105** — logs a request for ERT to action.
 - Dial-by-name 411** — spell a name to find an extension.
 - Missed-emergency follow-up** — callback/chase workflow so unanswered 111 calls are pursued, not lost.

DONE & VALIDATED LIVE

- SIP registration on real Android phones
- Two-way audio (198 echo, phone-to-phone)
- **111 rings a real answering handset** (on-shift, wallboard READY)
- Offline panic-coach (102)
- Real 111 recording captured
- Live status/control API (:8090)
- QEMU VM PBX, one-command deploy, autostart
- Full docs: 32 SOPs · 8 Blueprints · Journal

BUILDABLE NEXT (NO NEW HW)

- Real recorded voice prompts
- QR / profile phone provisioning
- Console served over the LAN
- Filled test-evidence sheet

DEFERRED BY DECISION

- DPDP / legal compliance — out of scope
- Cost / Bill of Materials — skipped

NEEDS HARDWARE / DECISION

- **#1 — move off QEMU SLIRP NAT** to a real LAN IP (van Linux box or bridged). Retires the whole media/NAT problem class.
- TLS + SRTP (encrypted signalling/media)
- Always-on ERT answer-point Androids
- Van UPS / power (Risk R1)
- mDNS / DHCP reservation across networks
- Full pilot + go-live sign-off

SET UP A NEW WINDOWS PC — ONE COMMAND

Installs prereqs, builds & boots the PBX VM, all accounts, autostart, Console + TV boards.

```
powershell -ExecutionPolicy Bypass -File .\Install-UpesEcs.ps1
```

Variants: `-LanIp 192.168.0.3` `-LaunchTV` · `-Rebuild` · `-Uninstall` . Internet needed only for the first build.

ADD A PERSON — IDEMPOTENT, NEVER CHANGES AN EXISTING PASSWORD

```
powershell -File deploy\qemu\Add-UpesUser.ps1 -SapId 500123394 -Name "Kinjal Srivastava"
```

SAP ID = extension = username. Generates the secret once, pins it as the single source of truth, pushes it live, syncs roster + callout groups + wallboard directory.

CAMPUS SCREENS & DAY-TO-DAY

Public safety board (giant DIAL 111)	:8080/tv-safety.html
Operations board (queue, KPIs)	:8080/tv-ops.html
Operations Console	localhost:8080
Launch both TVs (kiosk)	Show-TV.ps1 -Both
Re-point to current LAN IP	Set-UpesLanIp.ps1

REPOSITORY MAP

- **deploy/qemu/** — the real deployment (VM + lifecycle scripts)
- **deploy/asterisk/** — authoritative Asterisk config (accounts = source of truth)
- **api/ · Console/** — status API + console + wallboards
- **config/ · scripts/** — dialplan + in-VM helpers
- **provisioning/ · secrets/** — roster + SIP logins (never web-served)
- **SOP/ · Blueprint/ · Docs/** — plan, build, operate, decisions

Remaining work is execution, not design.

Stand up the server/van on a real LAN IP → run the Pilot Test Plan (SOP 17) → sign off go-live (SOP 18).

EMERGENCY

111